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New copper(II) complexes with dopamine hydrochloride and vanillymandelic acid: Spectroscopic and thermal characterization

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ABSTRACT

The dopamine derivatives participate in the regulation of wide variety of physiological functions in the human body and in medication life. Increase and/or decrease in the concentration of dopamine in human body reflect an indication for diseases such as Schizophrenia and/or Parkinson diseases. The Cu(II) chelates with coupled products of dopamine hydrochloride (DO.HCl) and vanillymandelic acid (VMA) with 4-aminoantipyrine (4-AAP) are prepared and characterized. Different physico-chemical techniques namely IR, magnetic and UV–vis spectra are used to investigate the structure of these chelates. Cu(II) forms 1:1 (Cu:DO) and 1:2 (Cu:VMA) chelates. DO behave as a uninegative tridentate ligand in binding to the Cu(II) ion while VMA behaves as a uninegative bidentate ligand. IR spectra show that the DO is coordinated to the Cu(II) ion in a tridentate manner with ONO donor sites of the phenolic–*OH*, –NH and carbonyl-*O*, while VMA is coordinated with OO donor sites of the phenolic–*OH* and –NH. Magnetic moment measurements reveal the presence of Cu(II) chelates in octahedral and square planar geometries with DO and VMA, respectively. The thermal decomposition of Cu(II) complexes is studied using thermogravimetric (TG) and differential thermal analysis (DTA) techniques. The activation thermodynamic parameters, such as, energy of activation, enthalpy, entropy and free energy change of the complexes are evaluated and the relative thermal stability of the complexes are discussed.

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1. Introduction

Hormones acted as chemical messengers controlling the activity of living things. In humans, they were released in small amounts in the blood stream from a number of vital sites especially some brain regions. Dopamine was recognized as neurotransmitter in its own right but the demonstration of its non-uniform distribution in the brain suggested that it might had a specific functional role of dopamine [1]. It had an important role in the pathogenesis of drug treatment of certain brain diseases, e.g. Parkinson's disease and Schizophrenia [2]. Parkinson's disease was associated with decrease of dopamine concentration in brain [3,4]. So, it must be medicated with drugs contain levodopa (LD) combined with certain amount of carbidopa (CD), which makes LD available for transport to the brain and converted to dopamine in the basal ganglia [5,6]. The brain of Schizophrenic patients produced more dopamine, especially in the striatum of the brain, than in normal brains [7]. In this case, the drug used block dopamine receptors in the brain. The increase of awareness of biological significance of dopamine derivatives has promoted a great deal of research into their detection and quantitative determination [8–10]. Several methods had been described for dopamine determination in biological samples and pharmaceutical formulations like chromatography and amperometry [11,12]. Miceller liquid chromatography with spectrophotometric procedures was applied to detect LD, 2-ethyldopa and soproterenol in their pharmaceutical preparations as well as other compounds frequently associated with them [13]. Many hospital laboratories were adopting HPLC method for measuring catecholamines [14–17]. However, HPLC technique consuming very time and restricted to highly specialized laboratories, so many spectrophotometric procedures were recommended for determination of dopamine using ninhydrin [18] and chloramine-T [19]. On screening literature, no studies concerning Cu(II) chelates of catecholamines coupled products with 4-AAP have been reported. Therefore, this work was undertaken in order to study the preparation and characterization of the solid chelates of dopamine hydrochloride and vanillymandelic acid and 4-AAP with Cu(II) tetramine complex. Interested solid products are separated and different physicochemical studies are used to infer the structure in order to give a clear idea about the role of metal ions on biological activity of these drugs in the biological systems. Thermal analyses (TG and DTA) techniques are also used to give a clear idea about the thermal behaviour and relative thermal stability of these chelates. The structures of the catecholamines used in this study are shown in Fig. 1.

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Fig. 1. Structural formulae of catecholamines and VMA.

2. Experimental

2.1. Materials

The chemicals used in this investigation were of the highest purity available. They included dopamine hydrochloride (DO.HCl), vanillylmandelic acid (VMA) and dobutamine hydrochloride (DOB.HCl) (supplied from Sigma), while levodopa (LD), carbidopa (CD), and 4-aminoantipyrine (4-AAP) were supplied from *Fluka Chemie AG*. Copper sulphate was supplied from *Sigma* while ammonia solution (33%) and sulphuric acid were supplied from *EL-Naser Pharmaceutical Chemical Co*. The water used was always de-ionized obtained from the instrument model *Altra Clear SG-Wasser aufbereitung und Regenerierststion GmbH*.

2.2. Equipments

The spectrophotometric measurements were carried out using the manual Spectronic 601 (Melton Roy Company), and Perkin-Elmer automated spectrophotometer UV–vis model Lambda 20 ranged from 200 to 900 nm. Measurements of pH were done using 716 DMS Titrino Metrohm connected with 728 Metrohm stirrer. For selecting accurate μL volumes, the calibrated transfer pipette (BRAND) was used in the range 100–1000 μL that transferred with disposable tips. Elemental analysis (C, H and N) were carried out at the Microanalytical Center, Cairo University. IR spectra were carried out using FTIR-8201 PC in the range 400–4000 cm^{-1}. The thermal analyses (TG and DTA) were carried out in dynamic nitrogen atmosphere (20 mL min^{-1}) with a heating rate of 10 °C min^{-1} using Shimadzu TG-50H and DTA-50H thermal analyzers. Mass spectra were carried out using MS-5988 mass spectrometer. The molar magnetic susceptibility was measured on powdered samples using the Faraday method. The diamagnetic corrections were made by Pascal's constant and $Hg[Co(SCN)_4]$ was used as a calibrant. The metal content of the complex (Cu) was determined by atomic absorption using Atomic Absorption Spectrophotometer Analyst 100 (Perkin Elmer) and manually using complexometric EDTA titration in a buffering medium of pH 11. The melting points of pure crystalline complexes were measured by Met-Temp (Laboratory Devices, USA) using Pyrex capillary melting tubes in the range 50–300 °C.

2.2.1. Solutions

5×10^{-3} mol L^{-1} solutions of DO.HCl, LD, α-MD, CD, AHT and VMA, and 2.5×10^{-3} mol L^{-1} solution of DOB-HCl were freshly prepared using de-ionized water, except DOB-HCl in methanol. 2×10^{-3} mol L^{-1} solutions of each catecholamines (DO.HCl, LD, α-MD, CD, AHT or VMA) or 3.38×10^{-4} mol L^{-1} solution of DOB.HCl were prepared by accurate dilution from the stock solution.

2.2.2. Procedures

2.2.2.1. Spectrophotometric determination of the dissociation constants. Ionization constants of catecholamines (DO.HCl, LD, α-MD, CD, AHT and DOB.HCl) and VMA were determined in aqueous medium using universal buffer of different pH values ranged from 2 to 12. 2×10^{-3} mol L^{-1} solutions of each of catecholamines (DO.HCl, LD, α-MD, CD, AHT or VMA) or 3.38×10^{-4} mol L^{-1} solution of DOB.HCl were completed to 10 mL in calibrated measuring flask using universal buffer of the desired pH value. The absorption spectra were scanned spontaneously in the wavelength range from 200 to 900 nm at room temperature. The absorbance was plotted as a function of pH from which the ionization constants were calculated using the recommended spectrophotometric methods [20]. Two spectrophotometric methods are applied to determine the dissociation constant values of the investigated compounds namely the half height and the limiting absorbance methods [20].

2.2.3. Absorption spectra

1×10^{-2} and 2×10^{-3} mol L^{-1} of DO.HCl and VMA were prepared, respectively, and complete absorption spectra were scanned in presence and absence of ammonia. On the other hand, complete absorption spectra for copper sulphate (1×10^{-2} mol L^{-1}) and 4-AAP (1×10^{-2} mol L^{-1}) were also done in presence and absence of ammonia solution. The absorption spectra were recorded for DO.HCl and VMA with copper sulphate or 4-AAP in presence of ammonia solution using the same concentrations as mentioned above. 1×10^{-2} mol L^{-1} of copper sulphate and 4-AAP solutions were mixed with 1×10^{-2} and 5×10^{-4} mol L^{-1} of DO.HCl and VMA, respectively, and the pH was adjusted to 10.0–11.0 using 0.35 mol L^{-1} ammonia solutions. The total volume in each case was completed to 10 mL in volumetric measuring flask using de-ionized water and the absorption spectra were recorded as mentioned above.

2.2.4. Determination of the stoichiometric ratio

2.2.4.1. Detection of most suitable ratio between copper sulphate: drug. The molar ratio method (mrm) described by Yoe and Jones [21] was applied in which the concentrations of copper sulphate (2×10^{-2} to 1×10^{-3} mmol) and 4-AAP (2×10^{-3} to 4×10^{-3} mmol) solutions were kept constant in the ratio 1:0.2 Cu(II):4-AAP. Ammonia solution was added to adjust the pH to 10.4. 0.05 mol L^{-1} solution of DO.HCl or VMA were added in different ratios ranged from 1:0.25 to 1:6 as [Cu(II)]:[DO] or [VMA] to the above prepared solution. The total volume of each mixture was completed up to 10 mL using de-ionized water. All mixtures were shacked well then the absorbances were recorded at 505, 500 and 480 nm after 30 and 33 min for DO.HCl or VMA, respectively.

2.2.4.2. Detection of most suitable ratio between 4-AAP: drug. Variable number of moles of 0.01 mol L^{-1} 4-AAP and DO.HCl or VMA was used with constant number of mmol from 0.01 mol L^{-1} copper sulphate solution. The ratio [Cu(II)]/(active ingredient]/[4-AAP]) was ranged from 1:0.5 to 1:6. All these concentrations were mixed in 10 mL measuring flask and the total volume was completed using de-ionized water. The procedure was completed as mentioned above.

2.2.5. Preparation of solid complexes

VMA (0.3964 g, 2 mmol) or DO.HCl (0.3792 g, 2 mmol) in de-ionized water (10 mL) was added dropwise to 4-AAP (0.4066 g, 2 mmol) in the same solvent, then copper tetramine complex (0.2497 g, 1 mmol) was added to them drop by drop. It was observed that deep bloody red colour was formed immediately after adding the copper tetramine complex. The excess solvents of water and ammonia solutions were evaporated whereupon red small crystals were formed. The solution was allowed to stand at room temperature for 24 h under dark conditions to give a chance for complete complex precipitation. The obtained crystals were filtered off from the supernatant using hersh funnel washed several times with water and then diethylether. Then the crystals were dried in vac-

uum dissecator at room temperature for few days. Finally this complex was collected and stored in dark and dry conditions.

3. Results and discussion

3.1. Absorption spectra of catecholamines

The absorption spectra of catecholamines under study in de-ionized water (DO.HCl, α-MD, CD, LD, AHT, and VMA), methanol (DOB.HCl) or in universal buffer of different pH values ranged from 2 to 12 are studied. It is clear that the absorption spectra of DO.HCl, α-MD, CD and DOB.HCl have λ_{max} at 280 nm ($\epsilon = 1.32 \times 10^4$, 2.21×10^4, 1.15×10^4 and 2.95×10^4 L mol^{-1} cm^{-1}, respectively), while LD, AHT, VMA and DOB.HCl have λ_{max} = 279 nm ($\epsilon = 1.39 \times 10^4$, 7.85×10^3 and 1.32×10^4 L mol^{-1} cm^{-1}, respectively).

Universal buffer of different pH values ranged from 2 to 12 was added to the active ingredients and scanning the absorption spectra from 200 to 900 nm. It is quietly obvious that from pH 2 to 10, the solution is colourless. From the data it is seen that the maximum absorbance at λ = 278–280 nm in de-ionized water is still in the same wavelength position in acidic range from pH 2 to 5. While from pH 6 to 10, the λ_{max} is shifted to be 279–287 nm. In case of VMA the λ_{max} at 279 nm is still in the same position as pH varied from 2 to 9 and shifted at pH 10 to be 285 nm ($\epsilon = 1.43 \times 10^4$ L mol^{-1} cm^{-1}). The red shift was obtained for all active ingredients at pH 11 and 12 to be λ = 286–298 nm, where the colour of the solutions changed from colourless to yellow. On the other hand, new maximum bands were appeared at pH 12 as a broad bands in the wavelength range from 428 to 432 nm ($\epsilon = 8.97 \times 10^3$ L mol^{-1} cm^{-1}) for DO.HCl and 438–442 nm for LD ($\epsilon = 8.85 \times 10^3$ L mol^{-1} cm^{-1}) and α-MD (3.5×10^3 L mol^{-1} cm^{-1}). The band observed at λ = 230 nm in case of VMA is not changed with the pH variation from 2 to 9. This band disappeared at pH = 10 and shifted to 250 nm at pH = 11 ($\epsilon = 4.39 \times 10^4$ L mol^{-1} cm^{-1}) and 12 ($\epsilon = 4.9 \times 10^4$ L mol^{-1} cm^{-1}). In case of DOB.HCl, the λ_{max} at 223 nm in de-ionized water was disappeared completely in using universal buffer of pH = 2–12. These bands can be attributed to n–π* and π–π* transitions within catecholamines molecules.

3.2. Determination of the dissociation constants (pK_a) using spectrophotometric methods

The present study is aimed to determine the dissociation constants of catecholamines under study. The spectra of these ligands have been studied in solutions of different pH values ranged from 2 to 12. The dissociation constants are calculated applying two spectrophotometric methods namely half height and limiting absorbance methods [20].

The protonation and deprotonation equilibria of LD, CD, α-MD, AHT, DO.HCl, VMA and DOB.HCl are suggested depending upon the influence of pH on the spectra of these compounds using universal buffer (pH = 2–12) applying the limiting absorbance method [20]. It is found that DO.HCl has two pK_a values at 6.3 and 10.8, while LD has three pK_a values at 4.2, 6.3 and 9.5. Meanwhile, α-MD, AHT and VMA have three pK_a values at 4.6, 6.8 and 10.3; 3.75, 6.1 and 10.1; and 2.8, 8.1 and 10.1, respectively. In case of CD and DOB.HCl, two pK_a values are obtained at 6.9 and 10.9; and 5.35 and 11.4, respectively.

The first pK_a value (pK_{a1} = 2.8–4.6) in case of LD, CD, α-MD, AHT and VMA may be attributed to the deprotonation of the carboxylic group. The second (pK_{a2} = 5.35–8.1) and the third (pK_{a3} = 9.5–11.4) dissociation constant values may be reasonably accounted for the deprotonation of the protonated amino and phenolic groups, respectively.

3.3. Determination of the stoichiometric ratio

Studying of the stoichiometric ratio between the active ingredients (DO.HCl and VMA) and reagents (copper tetramine or 4-AAP) reflects the most suitable ratio from reagents required for complete

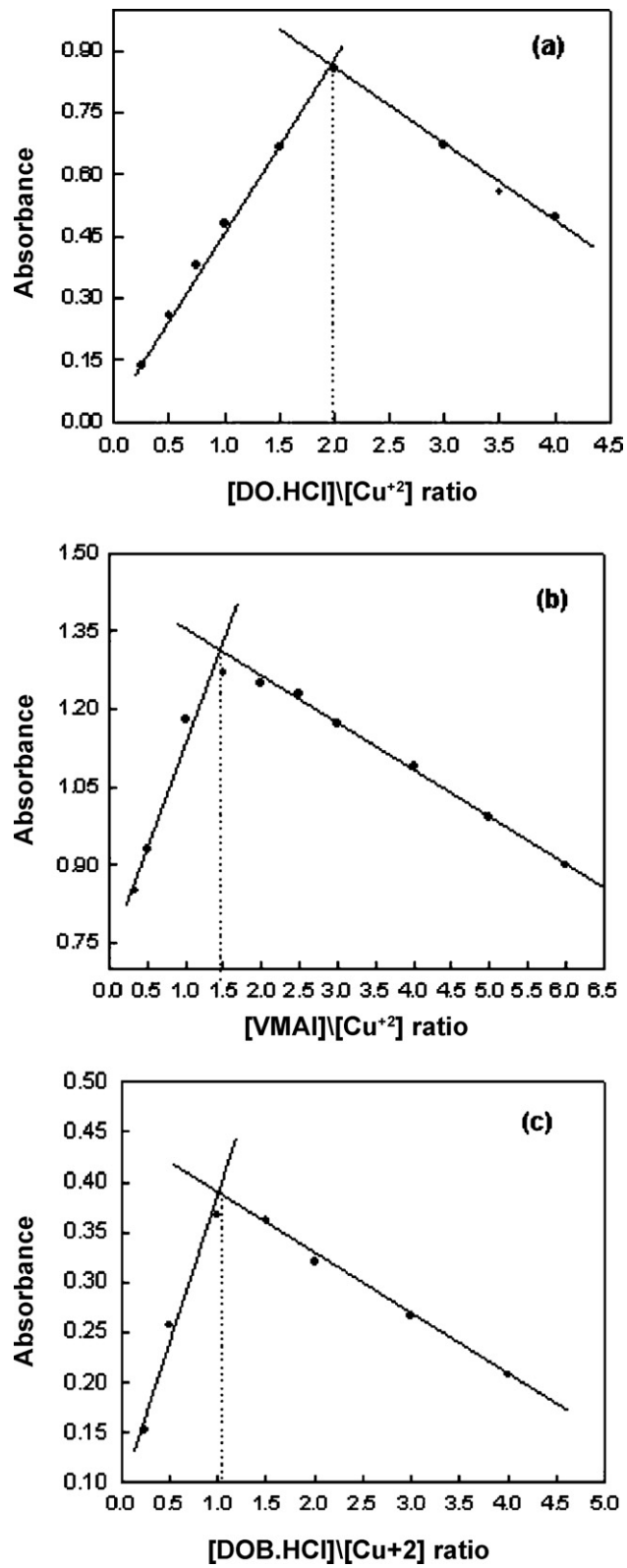


Fig. 2. Molar ratio of (a) [DO.HCl]/[Cu^{2+}] at λ = 505 nm, (b) [VMA]/[Cu^{2+}] at λ = 500 nm and (c) [DOB.HCl]/[Cu^{2+}] at λ = 480 nm.

complex formation. Molar ratio method [21] was applied for this purpose.

3.3.1. Determination of most suitable ratio of [Cu(II)]:[drug]

In order to find the suitable ratio, the reactants were added at different ratios ranged from 1:0.25 to 1:6 [Cu^{2+}]/[active ingredient]. From the plotted date in Fig. 2, it is obvious that the two extended lines are intersected at ratio 1:2 and 1:1.5 as [Cu^{2+}]/[DO.HCl] and [Cu(II)]/[VMA]], respectively, which is the most suitable ratio for complete complex formation.

3.3.2. Detection of most suitable ratio of [4-AAP]:[drug]

A study of the stoichiometric ratio between DO.HCl and VMA and 4-AAP reflects the most suitable ratio of 4-AAP that required for complete reaction formation. The results of the applied various [4-AAP]/[active ingredient] ratios ranged from 1:0.25 to 1:6 are shown in Fig. 3. On plotting the absorbance versus molar ratio; [4-AAP]:[drug], two straight lines are intercepted at 1:1 [4-AAP]/[active ingredient].

This can be explained in terms of formation of coupled dye product of 4-AAP and DO.HCl or VMA in 1:1 ratios then complex formation with Cu(II) tetramine complex in the ratio 1:2:2 or 1:1.5:1.5 [Cu(II)]/([DO.HCl]/[4-AAP]) or [Cu(II)]/([VMA]/[4-AAP]), respectively.

3.4. Effect of pH

Effect of pH on the reaction of different active ingredients was studied using ammonia solution as buffering media in the pH range from 9.5 to 12. The pH range from 9.5 to 10.0 is not a recommended media for determination of DO.HCl and VMA using copper sulphate and 4-AAP reagents, where uncompleted reactions occurred due to uncomplete formation of copper tetramine complex. The absorption spectra of 1×10^{-3} mol L^{-1} DO.HCl solution in different pH values using ammonia solution are studied. It is seen from this result that, as pH increase from 10.15 to 10.94, the absorbance decrease and the λ_{max} appears at 512 nm. By increasing the pH from 11.0 to 11.5, a blue shift in the λ_{max} to be at 505 nm is observed and the absorbance becomes stable. At pH = 12, the λ_{max} is highly blue shifted to be at 475 nm. At pH higher than 12, the colour and λ_{max} of the solution are completely disappeared.

In case of VMA, the λ_{max} appears as a broad band from 500 to 520 nm with pH 10.0 to 10.5. Deepening in colour of solution is observed and so the absorbance increases, then it becomes stable at pH from 10.1 to 10.3 and begins to decrease from pH 10.4 to 11.6. From pH 11.0, the lighting of solution colour is observed with the disappearance of λ_{max}.

3.5. Effect of time

The effect of time on the reaction between DO.HCl and VMA with copper tetramine complex and 4-AAP reagents was studied at λ_{max} = 505 and 500 nm, respectively, at different time intervals in order to choose the proper time needed for complete complex formation. In case of DO.HCl, as shown in Fig. 4a, during the first 25 min the absorbance decreases with the increase of time. As the time goes further, the absorbance decreases slowly up to 35 min where the absorbance is approximately stable. In case of VMA, Fig. 4b, the absorbance increases with time up to 30 min. From 35 to 40 min, the absorbance is approximately constant. The absorbance begins to increase slowly again up to 40 min. The effect of time on the reaction of DOB.HCl with copper tetramine and 4-AAP (Fig. 4c) shows that the absorbance decreases with time up to 15 min and the absorbance becomes stable for 30 min. After that the absorbance increases with slow rate.

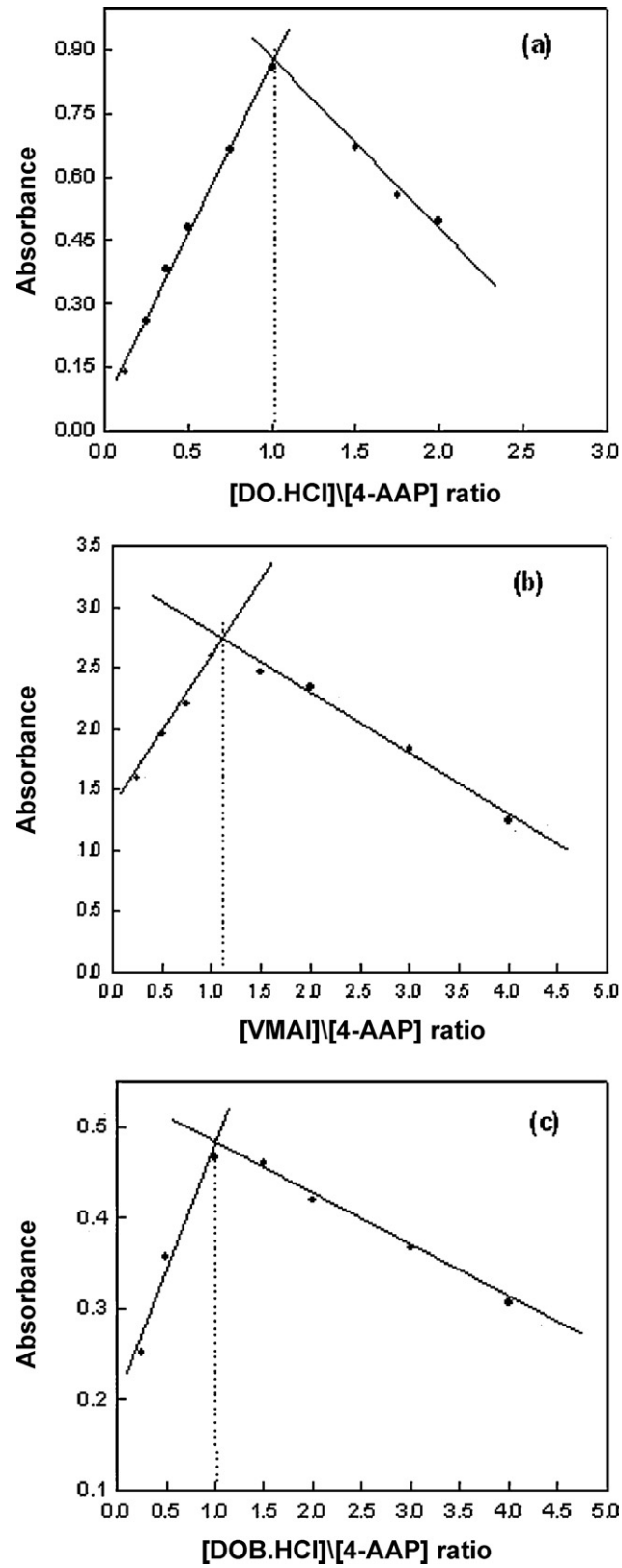


Fig. 3. Molar ratio of (a) [DO.HCl]/[4-AAP] at λ = 505 nm, (b) [VMA]/[4-AAP] at λ = 500 nm and (c) [DOB.HCl]/[4-AAP] at λ = 480 nm.

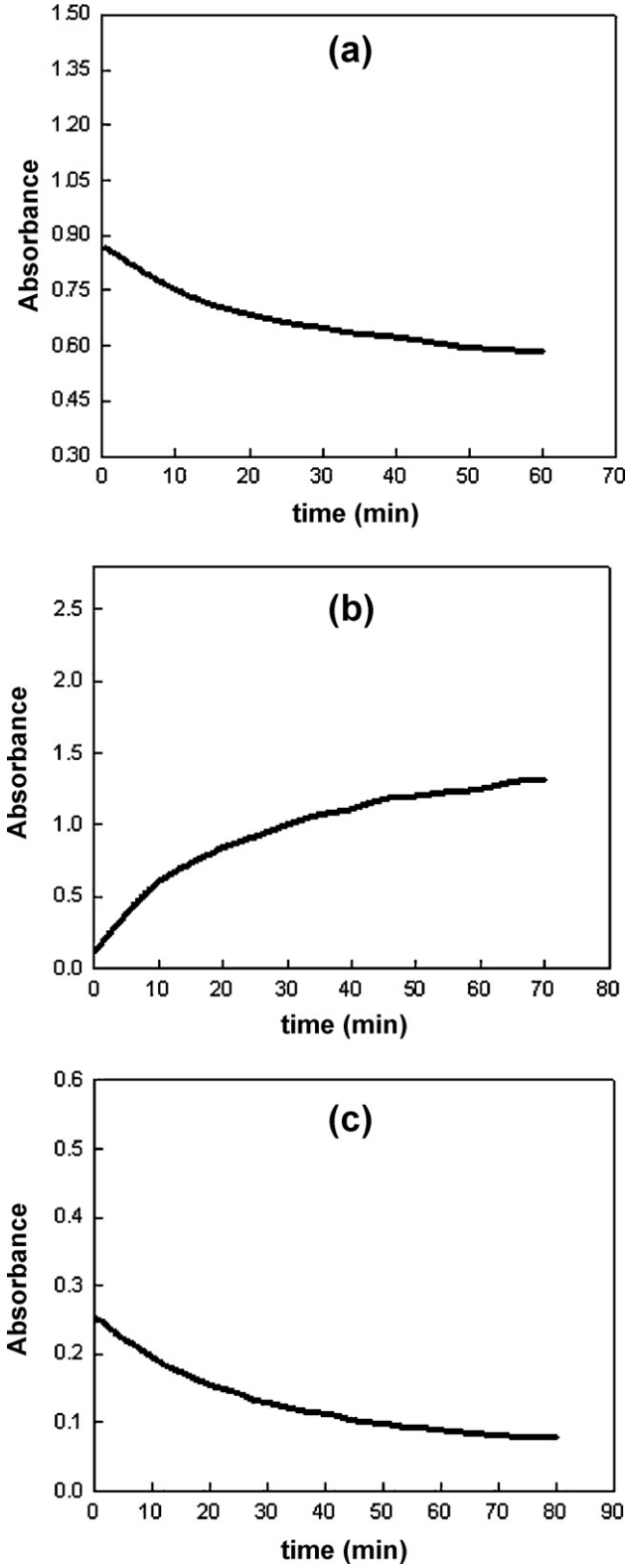


Fig. 4. Effect of time on the absorbance of (a) 0.01 M DO.HCl at λ = 505 nm, (b) 0.01 M VMA at λ = 500 nm and (c) 0.01 M DOB.HCl at λ = 480 nm using copper tetramine and 4-AAP reagents.

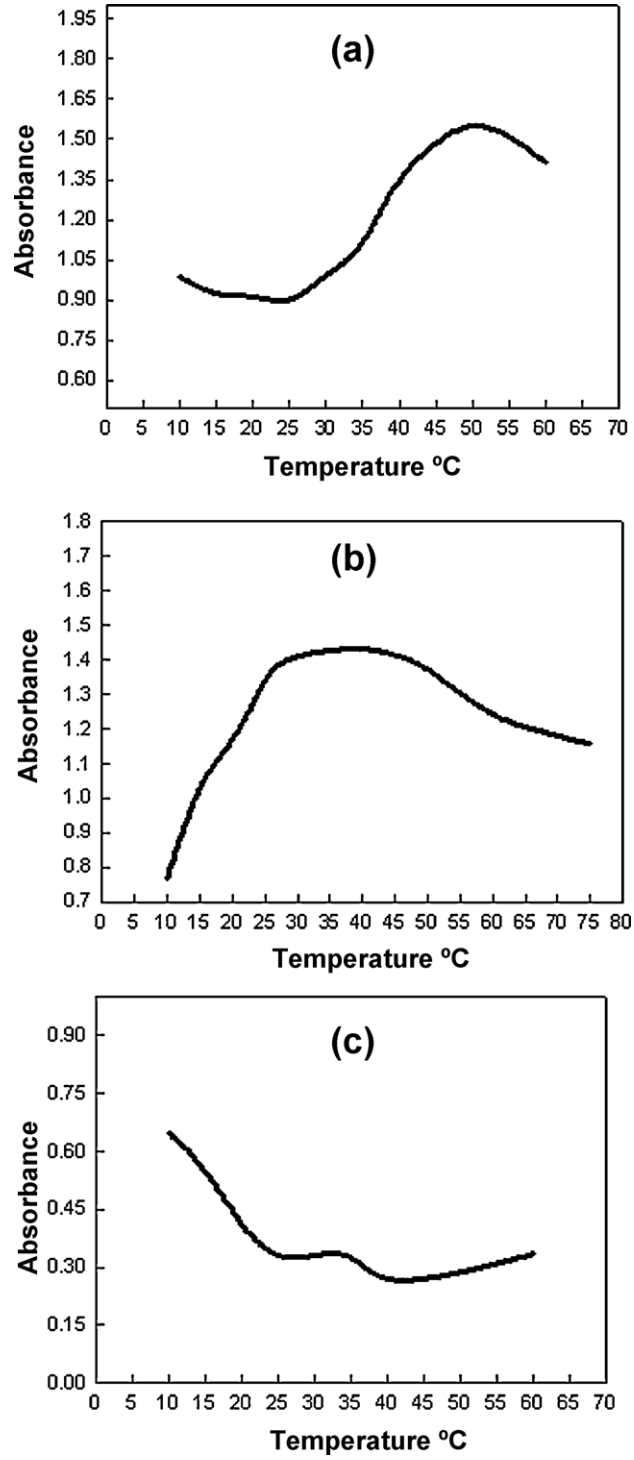


Fig. 5. Effect of temperature on the absorbance of (a) 0.01 M DO.HCl at λ = 505 nm, (b) 0.01 M VMA at λ = 500 nm and (c) 0.01 M DOB.HCl at λ = 480 nm using copper tetramine and 4-AAP reagents.

3.6. *Effect of temperature*

Due to the pronounced effect of temperature on the stability of the new coloured products that formed between DO.HCl and VMA with copper tetramine complex and 4-AAP reagents, the selection of optimum conditions is necessary. The effect of temperature on these reactions was studied in the temperature range from 10 to 60 °C (Fig. 5).

Table 1
Analytical and physical data of complexes A and B.

Complex	Colour (% yield)	M.p. (°C)	% Found (Calcd).				$\mu_{eff.}$ (B.M.)	Λ_m (Ω^{-1} mol^{-1} cm^2)
			C	H	N	M		
A $[Cu(C_{19}H_{21}N_4O_3)(OH)(H_2O)_2]\cdot 4½ H_2O$	Deep violet (62%)	>300 °C	41.79 (41.42)	5.83 (6.36)	10.68 (10.17)	11.53 (11.20)	2.03	22.30
B $[Cu(C_{20}H_{20}N_3O_6)_2]\cdot 2H_2O$	Deep violet (65%)	>300 °C	53.1 (53.6)	4.8 (4.7)	8.5 (9.7)	7.41 (7.09)	1.91	19.80

Fig. 5a shows the effect of temperature on the reaction of DO.HCl with copper tetramine and 4-AAP reagents. It is shown that, the absorbance is nearly constant in the temperature range from 15 to 25 °C, and horizontally increases with the increase of temperature up to 50 °C. This issue can be explained due to volatilization of ammonia at high temperatures, which leads to changing of pH medium. Via studying the effect of temperature variation on the reaction between VMA and copper tetramine complex and 4-AAP reagents (Fig. 5b), it is found that the absorbance increases as the temperature increases from 10 to 25 °C, then it is nearly constant from 30 to 40 °C. The final conclusion is that, 20 ± 2 and 27 ± 2 °C are the most suitable temperature values. In case of DOB.HCl, it is seen from Fig. 5c that, the absorbance decrease from 10 to 15 °C. From 20 to 35 °C, the absorbance is approximately constant. It is quietly obvious that, the absorbance decreases up to 40 °C then begin to increase with the temperature increase. Although higher temperatures give high absorbance values, lower temperatures have been chosen to avoid volatilization of ammonia at high temperature values, which affect complex formation.

3.7. *Composition and structure of the Cu(II) chelates with coupled products of DO.HCl and VMA with 4-AAP*

The analyzed solid products resulting from the reaction of DO.HCl and VMA with Cu(II) ion are prepared. The schemes of different reactions and proposed formulae of final products were suggested depending upon the data obtained from elemental analysis for C, H, N and metal content, IR spectra, thermal analysis (TG and DTA) and mass spectra. Analytical data for Cu(II) complexes with the coupled product of DO.HCl and VMA with 4-AAP are shown in Table 1. The elemental analyses results show that the calculated and found values of elements (C, H, N, and M) are close to each other and they are in the acceptable range of ±5% error giving rise to conclude that they are in good agreement with the proposed formulae.

It is seen from Table 1 that Cu(II) ion form complex with DO.HCl-4-AAP and VMA-4-AAP coupled products with molecular formulae $[Cu(C_{19}H_{21}N_4O_3)(H_2O)_2(OH)]\cdot 4½H_2O$ (A) and $[Cu(C_{20}H_{20}N_3O_6)_2]$ (B), respectively. The reaction of DO.HCl and VMA with 4-AAP in alkaline medium and hence the formation of coupled dye products may proceed as follows (Scheme 1) [22,23]:

The reaction of Cu(II) with HL proceeds as previously reported [22]:

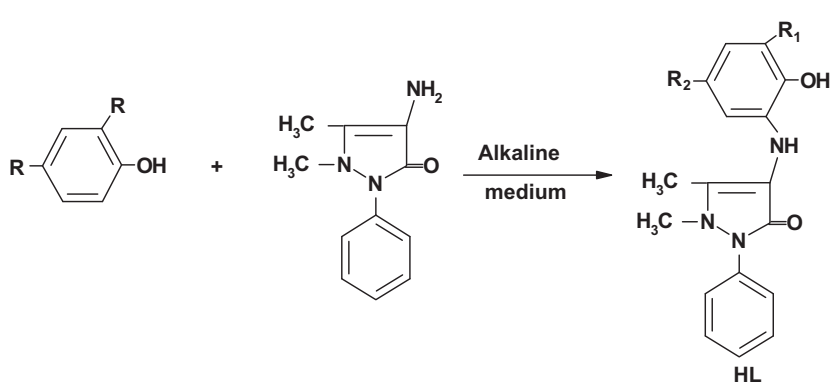
$$(A):\ Cu^{2+} + HL + OH^- + 6½\,H_2O \rightarrow [Cu(L)(H_2O)_2(OH)]\cdot 4½\,H_2O$$

$$(B):\ Cu^{2+} + 2HL \rightarrow [Cu(L)_2]$$

The complexes are coloured, stable toward light and the atmosphere and have high melting points. The isolated complexes are characterized and identified using different spectroscopic (IR and mass spectral measurements) magnetic and thermal analyses (TG and DTA) techniques. The molar conductance data of the complexes show their non electrolytic nature (Table 1). The room temperature magnetic moment values are found to be 2.03 and 1.91 BM for complex A and B, respectively. It reveals the presence of Cu(II) chelates in octahedral and square planar geometries with DO and VMA, respectively.

3.7.1. *IR spectral studies and mode of bonding*

In the absence of X-ray crystallography, IR spectroscopy technique is considered as powerful technique for elucidating the way of binding the organic ligands to the metal ions. The $\nu(NH_2)$ band at 3432 and 3327 cm^{-1} in the free 4-AAP ligand is disappeared and a weak band at 3143 and 3063 cm^{-1} in Cu(II) complexes with HL coupling ligands was appeared which indicate the replacement of the NH_2 proton during coupling with DO.HCl or VMA. This is sup-



For DO.HCl: R_1= OH , R_2= $CH_2\,CH_2\,NH_2$

For VMA: R_1= OCH_3 , R_2= CH(OH)(COOH)

Scheme 1. Formation of DO.HCl or VMA copper(II) 4-AAP coupled products.

Table 2
Thermoanalytical results (TG, DTG and DTA) of Cu complexes.

Complex	TG range (°C)	DTG_{max} (°C)	DTA (°C)	Weight loss, %		Total weight loss	Assignment
				Found	Calcd.		
A	30–350	82, 270	271 (−) 330 (+)	37.58	37.23	% Found = 62.47	Loss of 4½H_2O and $C_8H_{14}N$ molecules.
	350–800	470 730	428 (−) 510 (+) 602 (−)	29.6	28.7	% Calcd. = 65.93	Loss of $C_{11}H_{12}N$ molecule Leaving Cu_2O, Cu
B	25–140	57	53 (−)	3.87	4.02	% Found = 77.00	Loss of $2H_2O$
	140–950	255 373 750	311 (−) 450 (+) 720 (+)	75.65	75.04	% Calcd. = 79.06	Loss of $C_{20}H_{20}N$ molecule Leaving Cu_2O, Cu[225]

(−) Means endothermic reaction.
(+) Means exothermic reaction.

ported by the previously published data [23] and the previously results of spectrophotometric studies.

The IR spectra of the free DO.HCl and VMA ligands exhibit bands at 3345 and 3351 cm^{-1}, respectively, due to phenolic group stretching vibration (ν(OH)). The presence of hydrated water molecules renders it difficult to confirm the participation of this band in chelate formation. The shift of the bands due to ν(CO) and δ(OH) in-plane bending form 1266, 1392 and 1269, 1378 cm^{-1} in the free DO.HCl and VMA, respectively, to 1310 and 1408, and 1294 and 1398 cm^{-1} in Cu(II) complexes with DO.HCl and VMA coupled ligands, respectively, can be interpreted as proving the involvement of the phenolic oxygen atom in coordination [24,25].

The sharp IR band in the free 4-AAP ligand at 1648 cm^{-1} due to ν(C=O) is shifted to 1619 and 1654 cm^{-1} in the Cu(II) complexes with DO.HCl and VMA coupled ligands, respectively, confirming the involvement of the C=O oxygen in coordination [26]. Stretching vibration of the NH_2 group in the free DO.HCl in the region 3250–3100 cm^{-1} remain at about the same frequency in the spectrum of the Cu(II) complex, which indicates that the NH_2 group is slightly shared in coordination [27].

The bands due to ν(C=O), ν_{asym}(COO) and ν_{sym}(COO) located at 1747, 1609 and 1439 cm^{-1}, respectively, in the VMA ligand, are shifted to lower wavelength although it is not involved in complex formation. This shift may be attributed to the involvement of the carboxylate O atom in hydrogen bond with water of crystallization. The aquo complexes display a broad and medium band around 867–858 and 761–755 cm^{-1} which assigned to OH rocking and wagging vibration, respectively, indicating coordination of water in aquo complexes [28].

3.7.2. Thermal analyses (TG and DTA)

Thermal analyses techniques (TG and DTA) are useful in both quantitative and qualitative analyses. Samples can be identified and characterized by investigation of their thermal behaviour. TG measures weight changes, while DTA measures temperature of transitions and reactions.

The TG for Cu(II) complexes under study was carried out within the temperature range from room temperature up to 800 °C. The found mass losses were computed based on the TG results and the calculated mass losses were computed using the results of the microanalysis (Table 2).

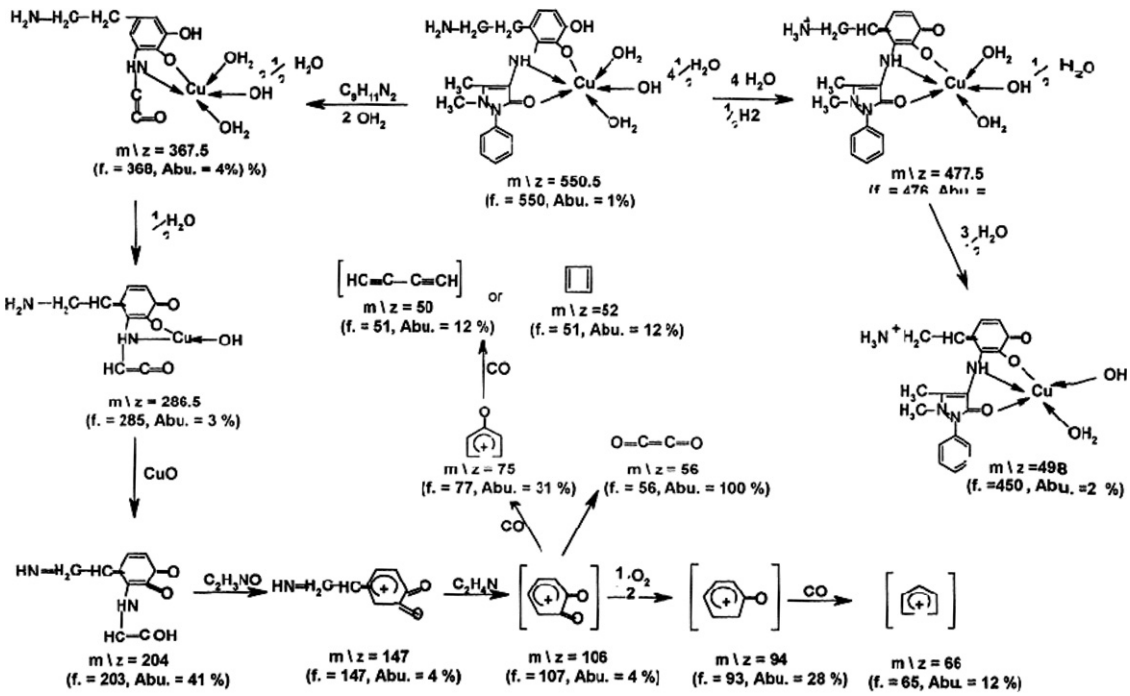
The TG curve for complex (A) shows that the complex decomposed in four steps. These steps extended between 30–350 and 350–800 °C. The first and second steps accompanied with weight loss 37.58% (calcd. 37.23%) which may be attributed to the loss of 4½ H_2O and $C_8H_{14}N$ molecule. The first two steps appear in DTA curve as endothermic peak at 271 °C and exothermic peak at 330 °C. The next two steps, which take place within the temperature range 350–800 °C, may involves the decomposition and loss of $C_{11}H_{12}N$ molecule with found weight loss 29.6% (calcd. = 28.7%). The later two steps appeared in DTA as endothermic peaks at 428 and 602 °C and exothermic peak at 510 °C.

The TG thermogram of complex (B) shows three steps of decomposition within the temperature range 25–800 °C. The first step within the temperature range from 25 to 140 °C corresponds to the loss of hydrated water with mass loss 4.02% (calcd. = 3.87%). The DTA curve shows one endothermic peak at 53 °C due to loss of this hydrated water. The next steps take place within the temperature range 140–950 °C with a mass loss 75.04% (calcd. = 75.65%). This mass may be attributed due to loss of $C_{20}H_{20}N$ molecule leaving Cu_2O and or Cu[225] as a residue. The DTA curve shows three peaks one endothermic at 255 °C and two exothermic at 373 and 750 °C, which may be due to pyrolysis of ligand. The appearance of many endothermic and exothermic peaks can be attributed to the gaseous decomposition of the water molecules anions and ligand molecules.

3.7.3. Mass spectra

The electron ionization (EI) mass spectra investigation of (A) and (B) complexes are studied which give valuable contributions about the structure of the fragmented ions. Schemes 2 and 3 have been proposed based on the basis of the mass measurements metastable ion studies and some high-resolution results. Fragmentation pattern of complex (A) shows the parent ion at m/z = 367.5. The first route involves mass fragmentation resulted in the formation of $[C_{10}H_{11}N_2O_4Cu]$ ion (M.wt = 286, m/z = 268.5, Abu. = 3%); $[C_{10}H_{11}N_2O_3]$ ion (M.wt = 204, m/z = 203, Abu. = 41%); $[C_8H_9NO_2]$ ion (M.wt = 147, m/z = 147, Abu. = 4%) and $[C_6H_5O_2]^+$ ion (M.wt = 106, m/z = 107, Abu. = 4%). This may be due to loss of ½ H_2O; CuO; C_2H_3NO and C_2H_4N, respectively. Further fragmentation gives $[C_5H_4O]^+$ ion (M.wt = 75, m/z = 77, Abu. = 31%); $[C_2O_2]$ ion as base peak (M.wt = 56, m/z = 56, Abu. = 100%); $[C_6H_5O]$ ion (M.wt = 93, m/z = 94, Abu. = 28%) and $[C_5H_5]^+$ ion (M.wt = 65, m/z = 65, Abu. = 12%). This may be due to loss of CO, ½ O_2 and CO, respectively. The second fragmentation route may involve the loss of hydrated water 4 H_2O and ½ H_2 followed by loss of 1½ H_2O to give $[C_{18}H_{25}N_4O_4Cu]^+$ ion (M.wt = 477, m/z = 476, Abu. = 21%) and $[C_{18}H_{22}N_4O_4Cu]^+$ ion (M.wt = 450, m/z = 448, Abu. = 2%) as shown in Scheme 2.

Fragmentation pattern of complex (B) (Scheme 3) represents two fragmentation routes. From Scheme 3, the first route involves successive mass fragmentation resulted in the formation of $[C_{19}H_{17}N_3O_6Cu]^+$ ion as base peak (M.wt = 445, m/z = 445.5, Abu. = 100%); due to loss of $C_{20}H_{20}N_3O_6$ and CH_3. The fragmentation continue by loss of (C_6H_5 and CO_2), $C_5H_7N_2O$, (CO and

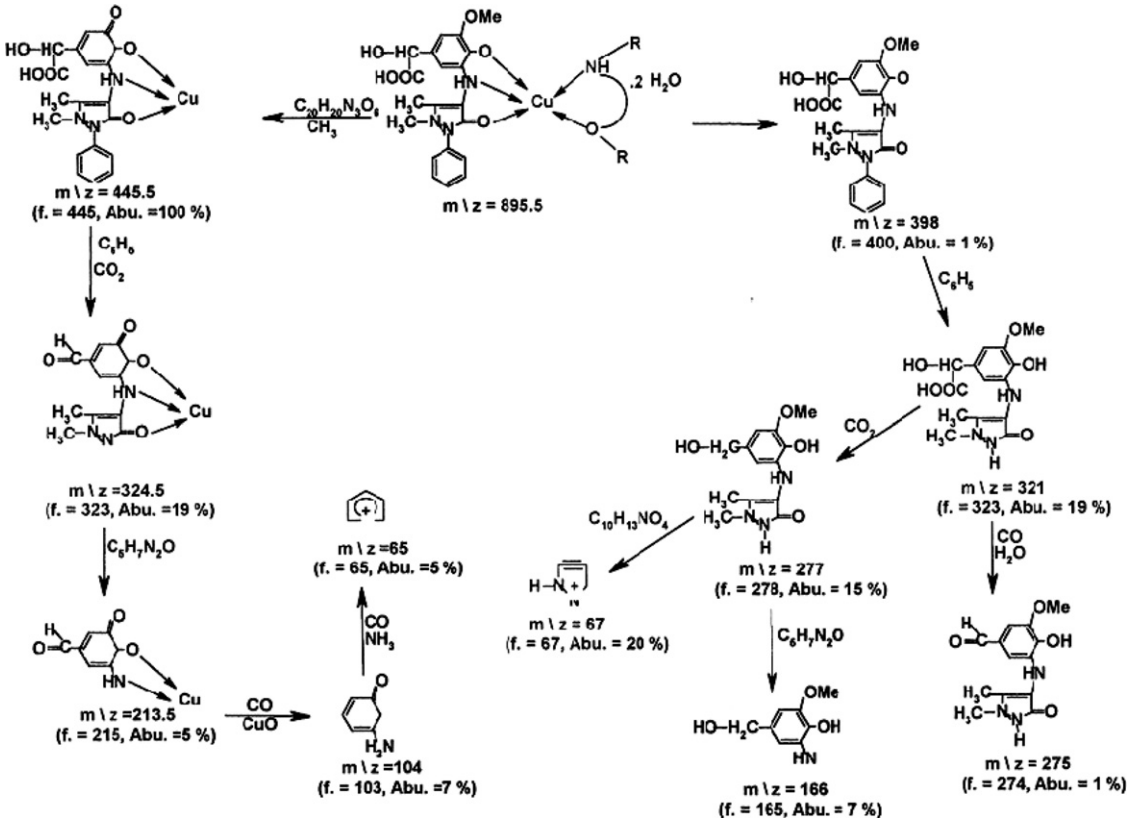


Scheme 2. Proposed mass fragmentation pathways of complex A.

CuO) and (NH_3 and CO) to form $[C_{12}H_{11}N_3O_4Cu]^+$ ion (M.wt = 323, m/z = 324.5, Abu. = 19%); $[C_7H_4NO_3Cu]$ ion (M.wt = 215, m/z = 213.5, Abu. = 5%); $[C_6H_4NO]$ ion (M.wt = 103, m/z = 104, Abu. = 7%) and $[C_5H_5]$ ion (M.wt = 65, m/z = 65, Abu. = 5%). The second fragmentation route can be explained by loss of C_6H_5, CO_2, and $C_{10}H_{12}NO_4$ to form $[C_{16}H_{17}N_3O_6]^+$ ion (M.wt = 323, m/z = 321, Abu. = 19%); $[C_{15}H_{17}N_3O_4]^+$ ion (M.wt = 278, m/z = 277, Abu. = 15%) and $[C_3H_2N_3]^+$ ion (M.wt = 67, m/z = 67, Abu. = 20%).

3.7.4. Structural interpretation

On the basis of the above observations, the structures of the complexes are shown in Fig. 6.



Scheme 3. Proposed mass fragmentation pathways of complex B.

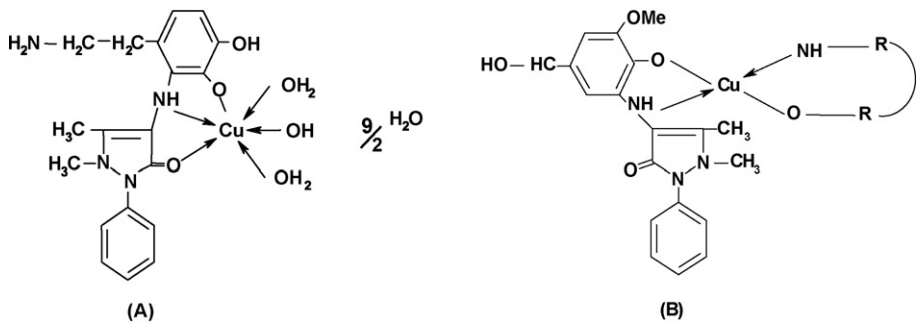


Fig. 6. Proposed structures of complexes (a) DO.HCl and (b) VMA with copper tetramine and 4 AAP.

4. Conclusion

The Cu(II) chelates with coupled products of dopamine hydrochloride (DO.HCl) and vanillymandelic acid (VMA) with 4-aminoantipyrine (4-AAP) are prepared and characterized. Different physico-chemical techniques namely IR, magnetic and UV–vis spectra are used to investigate the structure of these chelates. Cu(II) forms 1:1 (Cu:DO) and 1:2 (Cu:VMA) chelates as indicated by the molar ratio and continuous variation methods. The dissociation constants of the investigated catecholamines and vanillylmandelic acid were calculated spectrophotometrically. DO behave as a uninegative tridentate ligand in binding to the Cu(II) ion while VMA behaves as a uninegative bidentate ligand. IR spectra show that the DO is coordinated to the Cu(II) ion in a tridentate manner with ONO donor sites of the phenolic–*OH*, –NH and carbonyl-O, while VMA is coordinated with OO donor sites of the phenolic–*OH* and –NH. Magnetic moment measurements reveal the presence of Cu(II) chelates in octahedral and square planar geometries with DO and VMA, respectively. The thermal decomposition of Cu(II) complexes is studied using thermogravimetric (TG) and differential thermal analysis (DTA) techniques. The activation thermodynamic parameters, such as, energy of activation, enthalpy, entropy and free energy change of the complexes are evaluated and the relative thermal stability of the complexes are discussed.

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